

CertifEye AD CS ESC Audit Report

CertifEye - Active Directory Certificate Services defensive review

Prepared for	Current audit dataset
Date	2026-06-30
Scope	Scrubbed AD CS exports — issuance + posture; no authentication-use logs assumed
Data analyzed	17,250 certificate rows · 52 templates · 90 PKI ACEs · 1 CA row(s) · 11 DC row(s) · 4 web endpoint row(s) · 66 high-value token(s)
Overall posture	CRITICAL — one or more critical AD CS paths require validation/remediation

Read first. Issued-certificate evidence proves a certificate was *issued* — not that it was used to authenticate. Confirmation requires DC/Kerberos certificate-authentication logs. Posture findings describe conditions that can enable abuse, not proof that abuse occurred. Every remediation item is advisory and should be validated by a PKI administrator.

Executive Summary

Overall verdict

The highest-priority issues in this dataset are: F01-ESC5 (Critical; Template / PKI posture issue); F02-ESC1 (High; Likely ESC1 misuse); F03-ESC11 (High; Template / PKI posture issue). Treat issuance evidence as proof of issuance only until authentication-use logs confirm use.

Findings by severity and category

Severity	Critical	High	Medium	Low
Count	1	6	3	0

Category	Count
Template / PKI posture issue	7
Likely ESC1 misuse	1
Suspicious issuance needing validation	2

Top risks

- F01-ESC5 · Critical

2 dangerous PKI-object ACE(s) for PRINCIPAL_1a8fce5017e7babdcdbd920c3: object types CertificationAuthoritiesContainer, NTAAuthCertificates; dangerous rights include GenericAll, WriteDacl. Enroll-only rights were not counted as ESC5 control.

- F02-ESC1 · High

10 certificate issuance row(s) on template 'TPL_8a51cba45cc56d7160b059b2' have identity mismatch status DifferentMappedIdentity; 10 joined to template inventory, 10 are authentication-capable, and 0 involve HV/high-value tokens; first=2023-07-01, last=2025-04-18, revoked=0/10. Request IDs: REQ_0d584f2d4d82d5128f1ca45e;REQ_1effb448ce65d4e5f58979ca;REQ_331fb5da49982c6ba08b5931;REQ_52e68526ae1dec2d7d609721;REQ_68eb7786be2ceba9dcc9074;REQ_6b2f953a3acafea5773bae38;REQ_8a4c524373af052d1e2fb8a2;REQ_8fb39e7cad8f9ad2d46d9eb2.

- F03-ESC11 · High

CA security export indicates encrypted ICertRequest enforcement is not required for CA_2e1fd3acef6dc7f48d1e9451 (ESC11Candidate=True, IF_EnforceEncryptICertRequest=False).

- F04-ESC3 · High

6 certificate row(s) carry the Certificate Request Agent EKU and 6 on-behalf-of row(s) show caller/requester differences; high-value OBO rows: 0; first=2025-04-17, last=2026-04-17, revoked=0/6.

- F05-ESC4 · High

PRINCIPAL_db66978dcec117f0da16f56f has dangerous template-control rights on 1 template(s); 1 are published. Sample templates: TPL_4d61efba5e97e9618882ebec.

Prioritized recommendations

- **Validate ESC5 findings (1; F01-ESC5).** Confirm whether the principal is an approved PKI/tier-0 administrator and whether each listed ACE is intentional.

- **Remediate ESC5 findings (1; F01-ESC5).** Remove GenericAll/GenericWrite/WriteDacl/WriteOwner/WriteProperty from non-PKI-admin principals on the listed PKI objects; retain only explicit, documented PKI admin groups.

- **Validate ESC1 findings (1; F02-ESC1).** Correlate the certificate serials/thumbprints or RequestIDs with DC/Kerberos certificate-authentication logs before claiming confirmed misuse.

- **Remediate ESC1 findings (1; F02-ESC1).** If unexplained, revoke the certificates and tighten the issuing template's subject/SAN, approval, signature, and enrollment settings.

- **Validate ESC11 findings (1; F03-ESC11).** Confirm InterfaceFlags on the CA and whether ICertRequest encryption is enforced.

Remediation Guide

Every step below is advisory. Validate ownership and business need before making PKI changes.

ESC5

Loose permissions on core PKI directory objects

Severity: Critical · 1 finding(s)

What it is. Core Public Key Services objects determine trusted CAs, enrollment services, and available certificate templates.

How it can be abused. Write or ownership control can alter enterprise trust or PKI publication. Enroll-only rights and a CA host controlling its own CA object are not treated as control.

How to fix it.

1. Open ADSI Edit and navigate to Configuration -> CN=Public Key Services,CN=Services.
2. For each flagged object, open Properties -> Security and confirm the principal's approved role.
3. Remove GenericAll, GenericWrite, WriteDacl, WriteOwner, or equivalent control from non-PKI administrators and broad principals.
4. Retain the CA computer account on its own CA object only when required and documented.
5. Review parent-container inheritance so removed access is not reapplied.

How to verify. Re-export PKI object ACLs; confirm dangerous control ACEs for non-administrative principals are gone.

ESC1

Templates let requesters name the certificate identity

Severity: High · 2 finding(s)

What it is. A published authentication-capable template allows the requester to supply the subject or SAN.

How it can be abused. A requester can ask for a certificate naming a different account when enrollment scope and approval controls permit it.

How to fix it.

1. Open Certificate Templates (certtmpl.msc) and identify every flagged published template.
2. On Subject Name, select Build from this Active Directory information unless requester-supplied identity is required.
3. If requester input is required, enable CA manager approval or require authorized signatures.
4. On Security, restrict Enroll and Autoenroll to the smallest approved group; remove broad/default enrollment.
5. Revoke and reissue affected certificates when validation confirms inappropriate issuance.

How to verify. Re-export templates and issuance. Confirm requester-supplied subject/SAN is disabled or approval/signature controls are present, and review remaining mismatches.

ESC11**Certificate requests must require encrypted RPC**

Severity: High · 1 finding(s)

What it is. The CA RPC interface may accept certificate requests without packet privacy when IF_ENFORCEENCRYPTICERTREQUEST is not set.

How it can be abused. Unprotected request traffic can satisfy a relay prerequisite depending on other environmental controls.

How to fix it.

1. Confirm InterfaceFlags live on every issuing CA and identify legacy clients that may be affected.
2. Enable IF_ENFORCEENCRYPTICERTREQUEST during an approved maintenance window.
3. Restart Certificate Services and validate approved enrollment clients.
4. Pair the change with NTLM reduction, SMB/LDAP hardening, and monitoring of failed enrollment requests.

How to verify. Re-read InterfaceFlags and confirm the CA rejects enrollment requests that do not provide the required RPC protection.

ESC3**Enrollment-agent and on-behalf-of issuance needs tight restriction**

Severity: High · 1 finding(s)

What it is. Enrollment Agent capability permits approved identities to request certificates on behalf of others.

How it can be abused. An over-broad agent population or missing agent restrictions can allow unauthorized delegated issuance.

How to fix it.

1. Inventory every Enrollment Agent certificate and validate its owner, purpose, validity, and revocation state.
2. Configure CA Enrollment Agent restrictions to limit which agents may use which templates for which subjects.
3. Restrict Enrollment Agent template enrollment and require authorized signatures where appropriate.
4. Remove stale agents and revoke certificates that no longer have an approved owner or purpose.

How to verify. Re-export agent and on-behalf-of issuance; confirm every agent is approved and CA restrictions cover its permitted templates and subjects.

ESC4**Non-PKI principals can modify certificate templates**

Severity: High · 1 finding(s)

What it is. Template-control rights can allow a principal to reshape a template into a higher-risk configuration.

How it can be abused. Control over publication, subject-name behavior, EKUs, approval, or security can create an unsafe enrollment path.

How to fix it.

1. Open each flagged template in certtmpl.msc and review its Security descriptor.
2. Remove GenericAll, GenericWrite, WriteDacl, WriteOwner, WriteProperty, and Full Control from non-PKI administrators.
3. Use a dedicated, documented PKI administration group for template changes.
4. Review inherited permissions and the parent Certificate Templates container for the source of access.
5. Compare the final template settings with the approved baseline before republishing.

How to verify. Re-export template ACLs; confirm only approved PKI administrators retain modification rights.

ESC8

Web enrollment authentication and EPA need hardening

Severity: High · 3 finding(s)

What it is. AD CS web endpoints may expose Windows authentication over HTTP or HTTPS without confirmed Extended Protection.

How it can be abused. An exposed NTLM-capable endpoint can satisfy a relay prerequisite when transport and EPA protections are absent.

How to fix it.

1. Inventory every certsrv, CES, and CEP endpoint and confirm whether it is still required.
2. Disable HTTP bindings; require HTTPS with a valid certificate and redirect or remove legacy listeners.
3. Disable NTLM where feasible and require Extended Protection for Authentication on Windows-authenticated endpoints.
4. Restrict network reachability to approved enrollment clients and monitor endpoint authentication.
5. Retire unused web enrollment roles instead of leaving them partially configured.

How to verify. Recollect endpoint evidence and confirm HTTP is absent, NTLM exposure is reduced, and EPA is Required or otherwise explicitly validated.

ESC9/10

Certificate-to-account mapping needs full enforcement

Severity: High · 1 finding(s)

What it is. Certificates missing or mismatching the SID security extension require strong domain-controller mapping enforcement.

How it can be abused. Weak mapping modes may accept a certificate for an unintended account; unreadable or intended-only policy does not prove protection.

How to fix it.

1. Confirm StrongCertificateBindingEnforcement live on every writable and read-only domain controller.
2. Move all DCs to Full enforcement after completing Microsoft's compatibility review.
3. Identify and reissue certificates that lack the SID security extension where the template should include it.
4. Investigate actionable identity mismatches and document benign mapping cases separately.
5. Monitor certificate-authentication events during and after enforcement rollout.

How to verify. Recollect live DC values and confirm every readable DC reports Full enforcement; validate remaining certificate mappings through approved authentication testing.

Technical Analysis

Methodology

The analysis uses explicit field checks over the supplied scrubbed CSVs. Files are matched by schema and filename hints, not by one fixed client export name. Missing files lower coverage rather than causing clean claims. Counts, categories, and findings are computed programmatically from the current data only.

Data quality & coverage

Certificate rows	17,250; ParseStatus OK: 17,250; Issued: 17,227; Revoked: 15
Identity mapping	present
IdentityMappingStatus	NoSANUPN 12930, SameIdentity 4310, DifferentMappedIdentity 10
Template join	Joined 17250
Issuance signals	ESC6 SAN-attr 0 · ESC3 agent 6 · ESC3 OBO 6 · ESC2 0 · SID missing 7 · SID mismatch real 7
Files supplied	certs=yes, templates=yes, acs=yes, ca=yes, dc=yes, web=yes, hv=yes
DC strong mapping	Full enforcement 6/11; live reads 10; methods RemoteRegistry, WinRM

ESC coverage matrix

ESC	Status	Evidence
ESC1	High	1 ESC1-shaped published template(s); 10 requester/SAN mismatch issuance row(s).
ESC2	Clean	0 Any Purpose/no-EKU issued certificate row(s).
ESC3	High	6 enrollment-agent certificate row(s); 6 on-behalf-of row(s).
ESC4	High	1 template(s) flagged with dangerous template-control rights.
ESC5	Critical	2 write/control PKI ACE(s) after excluding enroll-only and CA-host self-control rows.
ESC6	Clean	CA flag true on 0 CA row(s); 0 strong issuance-side SAN-attribute indicator row(s).
ESC7	Clean	0 CA row(s) marked ESC7Candidate.
ESC8	High	4 endpoint row(s); confirmed=0, metadata risk=1, needs EPA check=3.
ESC9/10	High	SID signals=14; DC Full enforcement 6/10 readable row(s); unreadable=1.
ESC11	High	1 CA row(s) marked ESC11Candidate or not enforcing encrypted ICertRequest.

Controls verified clean / mitigated

- ESC2 - 0 Any Purpose/no-EKU issued certificate row(s).
- ESC6 - CA flag true on 0 CA row(s); 0 strong issuance-side SAN-attribute indicator row(s).
- ESC7 - 0 CA row(s) marked ESC7Candidate.

Findings — Likely ESC1 misuse

Finding	Sev	Conf	Evidence
F02-ESC1	High	High	10 certificate issuance row(s) on template 'TPL_8a51cba45cc56d7160b059b2' have identity mismatch status DifferentMappedIdentity; 10 joined to template inventory, 10 are authentication-capable, and 0 involve HV/high-value tokens; first=2023-07-01, last=2025-04-18, revoked=0/10. Request IDs: REQ_0d584f2d4d82d5128f1ca45e;REQ_1effb448ce65d4e5f58979ca;REQ_331fb5da49982c6ba08b5931;REQ_52e68526ae1dec2d7d609721;REQ_68eb7786be2ceba9dcc9074;REQ_6b2f953a3acafea5773bae38;REQ_8a4c524373af052d1e2fb8a2;REQ_8fb39e7cad8f9ad2d46d9eb2.

Findings — Suspicious issuance needing validation

Finding	Sev	Conf	Evidence
F04-ESC3	High	High	6 certificate row(s) carry the Certificate Request Agent EKU and 6 on-behalf-of row(s) show caller/requester differences; high-value OBO rows: 0; first=2025-04-17, last=2026-04-17, revoked=0/6.
F07-ESC9-10	High	Medium	SID-extension signals: missing=7, real mismatches=7. 6/10 readable DC row(s) show Full enforcement; unreadable=1.

Findings — Template / PKI posture issue

Finding	Sev	Conf	Evidence
F01-ESC5	Critical	High	2 dangerous PKI-object ACE(s) for PRINCIPAL_1a8fce5017e7babdcdbd920c3: object types CertificationAuthoritiesContainer, NTAuthCertificates; dangerous rights include GenericAll, WriteDacl. Enroll-only rights were not counted as ESC5 control.
F03-ESC11	High	High	CA security export indicates encrypted ICertRequest enforcement is not required for CA_2e1fd3acef6dc7f48d1e9451 (ESC11Candidate=True, IF_EnforceEncryptICertRequest=False).
F05-ESC4	High	High	PRINCIPAL_db66978dcec117f0da16f56f has dangerous template-control rights on 1 template(s); 1 are published. Sample templates: TPL_4d61efba5e97e9618882ebec.
F06-ESC8	High	High	Web enrollment endpoint COMPUTER_9482814825d3dc360e819cd2: Scheme=http, IsHttp=True, NTLM offered=True, EPA=None, ESC8Confirmed=False, NeedsEpaCheck=False.
F08-ESC1	Medium	High	1 published template(s) are ESC1-shaped: requester-supplied Subject/SAN, authentication-capable EKU or Any Purpose/no EKU, no manager approval, no authorized signatures, and enrollment rights present. Broad enrollment templates: 0.
F09-ESC8	Medium	Medium	Web enrollment endpoint COMPUTER_7034fce70867a0dac7048921: Scheme=https, IsHttp=False, NTLM offered=unknown, EPA=Unknown, ESC8Confirmed=False, NeedsEpaCheck=True.
F10-ESC8	Medium	Medium	Web enrollment endpoint COMPUTER_507b264ef7a8e6eb273ef341: Scheme=https, IsHttp=False, NTLM offered=True, EPA=Unknown, ESC8Confirmed=False, NeedsEpaCheck=True.

Evidence trail — High & Critical findings

F01-ESC5 · Critical

Evidence: 2 dangerous PKI-object ACE(s) for PRINCIPAL_1a8fce5017e7babdcdbd920c3: object types CertificationAuthoritiesContainer, NTAuthCertificates; dangerous rights include GenericAll, WriteDacl. Enroll-only rights were not counted as ESC5 control.

Why it matters: Write/control rights over PKI AD objects can let that principal alter certificate trust, publish or modify templates, or change enrollment-service objects. That is a PKI privilege-escalation path if the principal is not an intended PKI administrator.

Validation: Confirm whether the principal is an approved PKI/tier-0 administrator and whether each listed ACE is intentional.

Remediation: Remove GenericAll/GenericWrite/WriteDacl/WriteOwner/WriteProperty from non-PKI-admin principals on the listed PKI objects; retain only explicit, documented PKI admin groups.

F02-ESC1 · High

Evidence: 10 certificate issuance row(s) on template 'TPL_8a51cba45cc56d7160b059b2' have identity mismatch status DifferentMappedIdentity; 10 joined to template inventory, 10 are authentication-capable, and 0 involve HV/high-value tokens; first=2023-07-01, last=2025-04-18, revoked=0/10. Request IDs: REQ_0d584f2d4d82d5128f1ca45e;REQ_1effb448ce65d4e5f58979ca;REQ_Q_331fb5da49982c6ba08b5931;REQ_52e68526ae1dec2d7d609721;REQ_68eb7786be2ceba9dccb9074;REQ_6b2f953a3acafea5773bae38;REQ_8a4c524373af052d1e2fb8a2;REQ_8fb39e7cad8f9ad2d46d9eb2.

Why it matters: Requester/SAN identity mismatch on an authentication-capable certificate is the core signal for certificate-based impersonation. Issuance alone does not prove the certificate was used.

Validation: Correlate the certificate serials/thumbprints or RequestIDs with DC/Kerberos certificate-authentication logs before claiming confirmed misuse.

Remediation: If unexplained, revoke the certificates and tighten the issuing template's subject/SAN, approval, signature, and enrollment settings.

F03-ESC11 · High

Evidence: CA security export indicates encrypted ICertRequest enforcement is not required for CA_2e1fd3acef6dc7f48d1e9451 (ESC11Candidate=True, IF_EnforceEncryptICertRequest=False).

Why it matters: If request traffic is not protected, NTLM relay paths to the CA RPC interface may be possible depending on environment controls.

Validation: Confirm InterfaceFlags on the CA and whether ICertRequest encryption is enforced.

Remediation: Enable encrypted ICertRequest enforcement and restart the CA service after validation.

F04-ESC3 · High

Evidence: 6 certificate row(s) carry the Certificate Request Agent ECU and 6 on-behalf-of row(s) show caller/requester differences; high-value OBO rows: 0; first=2025-04-17, last=2026-04-17, revoked=0/6.

Why it matters: Enrollment Agent capability can be legitimate, but it can also enable requesting certificates on behalf of other principals if not tightly restricted.

Validation: Confirm all enrollment agents are authorized and that CA Enrollment Agent restrictions limit which templates and principals they can request for.

Remediation: Restrict enrollment-agent template enrollment, configure CA Enrollment Agent restrictions, and require authorized signatures on sensitive templates.

F05-ESC4 · High

Evidence: PRINCIPAL_db66978dcec117f0da16f56f has dangerous template-control rights on 1 template(s); 1 are published. Sample templates: TPL_4d61efba5e97e9618882ebee.

Why it matters: A principal that can modify a certificate template can potentially reshape it into an ESC1-capable template and then request authentication-capable certificates.

Validation: Confirm whether the principal is an intended template/PKI administrator. Recurring control across many templates may be normal PKI administration, but should be explicitly validated.

Remediation: Remove Write/Full Control/WriteDacl/WriteOwner from non-PKI-admin principals on certificate templates; use a dedicated PKI admin group.

F06-ESC8 · High

Evidence: Web enrollment endpoint COMPUTER_9482814825d3dc360e819cd2: Scheme=http, IsHttp=True, NTLM offered=True, EPA=None, ESC8Confirmed=False, NeedsEpaCheck=False.

Why it matters: AD CS web enrollment can be exposed to NTLM relay risk if HTTP/NTLM is available and EPA/channel binding is absent or unverified.

Validation: Confirm endpoint reachability, authentication schemes, and IIS Extended Protection settings from the web host.

Remediation: Require HTTPS, disable NTLM where possible, require Extended Protection for Authentication, or disable web enrollment if unused.

F07-ESC9-10 • High

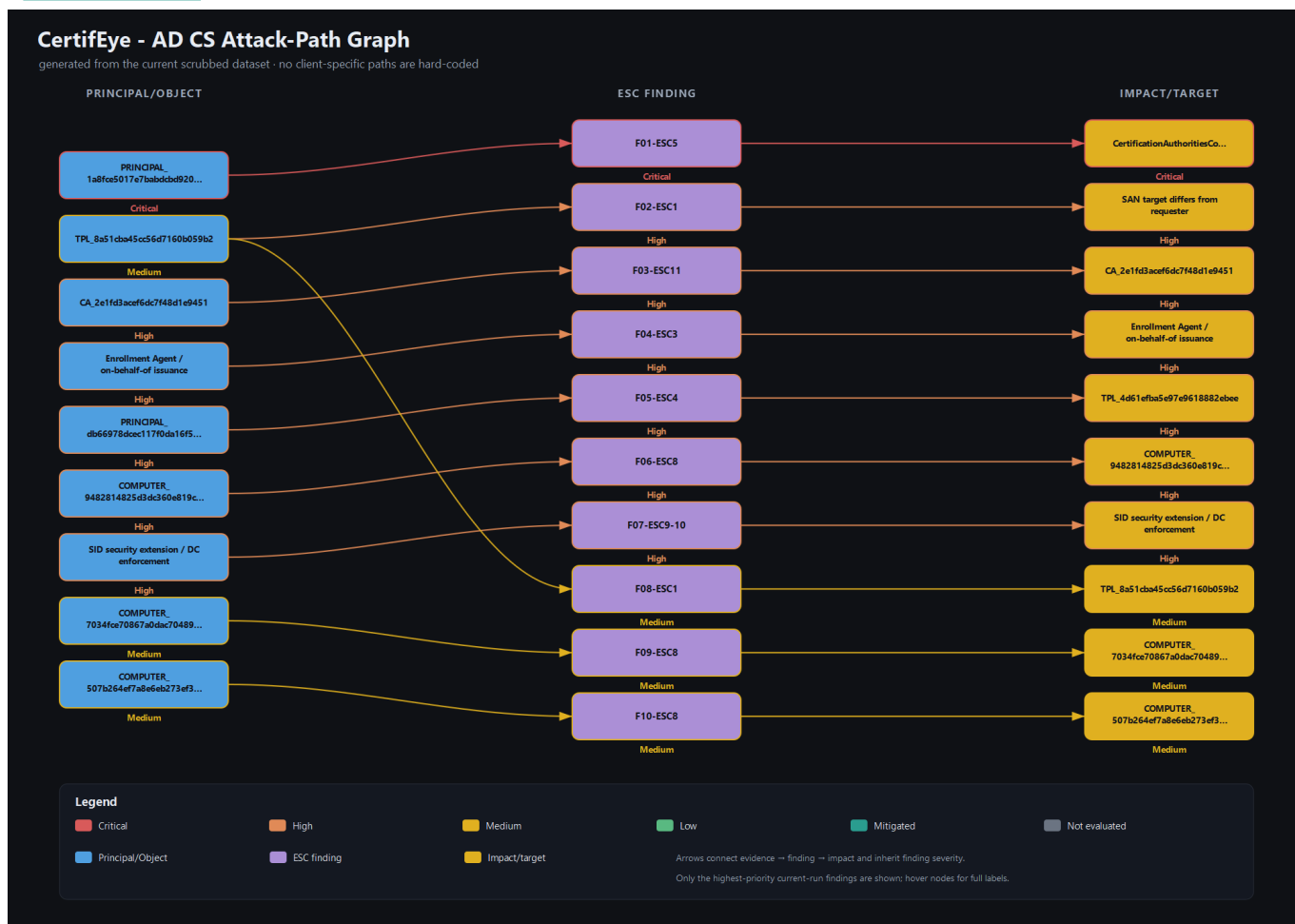
Evidence: SID-extension signals: missing=7, real mismatches=7. 6/10 readable DC row(s) show Full enforcement; unreadable=1.

Why it matters: Weak certificate mapping is exploitable only when domain controllers accept weak mappings for certificates missing or mismatching the SID security extension.

Validation: Confirm StrongCertificateBindingEnforcement live on every DC and correlate any suspicious certificates to authentication logs.

Remediation: Set StrongCertificateBindingEnforcement=2 on all DCs and reissue certificates with the SID security extension where applicable.

Attack-path graph



Appendix — assumptions, token model, limitations

- All inputs are assumed scrubbed. Tokens such as PRINCIPAL_x, HV_PRINCIPAL_x, GROUP_x, COMPUTER_x, TEMPLATE_x, CA_x, and X500_x are anonymized; broad/default labels are intentionally preserved for risk scoring.
- No real identities were inferred or reverse-mapped.
- Issued-certificate evidence proves issuance, not authentication or use. Confirmation requires DC/Kerberos certificate-authentication logs.
- If an ESC area is marked Not evaluated, the corresponding input file was not supplied or did not contain recognizable columns.
- All remediation is advisory guidance for a PKI administrator to validate and apply in a maintenance window.